



Date: June 3, 2025

Trimester 3
Practice Test

Name:

Class of:

Teacher:

Homerroom:

Part 1: Multiple Choice

1. The value $V(t)$ of an appliance depreciates according to

$$V(t) = P(0.90)^t,$$

where P is the purchase price and t is the number of years since purchase. By what percentage does the appliance depreciate each year?

Solution. Because the value is multiplied by 0.90 each year, the appliance keeps 90% of its value and therefore loses $100\% - 90\% = 10\%$ per year.

10%

2. Select all of the following that are equivalent to $\log_5 50$.

- A. $\log_5 10 + \log_5 5$
- B. $\log_5 100 - \log_5 2$
- C. $5 \log_5 2$
- D. $-\log_5\left(\frac{1}{50}\right)$

Solution.

A: $\log_5(10) + \log_5(5) = \log_5(50) \checkmark$

B: $\log_5(100) - \log_5(2) = \log_5\left(\frac{100}{2}\right) = \log_5(50) \checkmark$

C: $5 \log_5 2 = \log_5(2^5) = \log_5(32) \times$

D: $-\log_5\left(\frac{1}{50}\right) = \log_5(50) \checkmark$

A, B, D

3. At what x -value do the graphs $y = 4^{x-1}$ and $y = \left(\frac{1}{2}\right)^3$ intersect?

- A. $x = -\frac{1}{2}$
- B. $x = \log_4\left(\frac{1}{8}\right) + 1$
- C. $x = -\log_2 8$
- D. $x = \log_{\frac{1}{2}}(8) - 1$

Solution.

$$4^{x-1} = \left(\frac{1}{2}\right)^3 \implies 2^{2x-2} = 2^{-3}$$

$$2x - 2 = -3 \implies x = -\frac{1}{2}$$

$x = -\frac{1}{2}$

 (choice A)

4. Splash City is building a pool with a 3-ft-wide walkway. The pool's width is 15 ft less than its length L . The outer (walkway + pool) area is 2400 ft^2 . Which quadratic in standard form models the area?

- A. $L^2 - 3L - 2454 = 0$
- B. $L^2 - 15L - 54 = 0$
- C. $L^2 + 6L - 2400 = 0$
- D. $L^2 - 9L - 2400 = 0$

Solution.

$$\text{Outer length} = L + 2(3) = L + 6, \quad \text{Outer width} = L - 15 + 2(3) = L - 9$$

$$(L + 6)(L - 9) = 2400 \implies L^2 - 3L - 2454 = 0$$

A

5. Two horizontal parallel lines are cut by two negative-sloping parallel transversals, forming a quadrilateral. Select the true statement.

- A. The angles on the same vertical line are congruent.
- B. The angles on the same horizontal line are supplementary.
- C. The opposite interior angles are congruent.
- D. No interior angles are congruent.

Solution. The setup describes a quadrilateral formed by:

- Two horizontal parallel lines, and
- Two parallel transversals sloping downward (negative slope).

This forms a parallelogram, because both pairs of opposite sides are parallel.

In any parallelogram:

- Opposite interior angles are congruent. (True)
- Consecutive interior angles are supplementary (not necessarily on a horizontal line). (Misleading)
- Angles on a vertical line (called vertical angles) can be congruent, but not all such angles in this diagram are formed by intersecting lines. (False/Irrelevant)
- The claim that no interior angles are congruent contradicts the property of a parallelogram. (False)

C. The opposite interior angles are congruent.

6. Let r be irrational. Which of the following are always irrational? (select all)

- A. r^3
- B. $r + 2$
- C. $5r$
- D. $\frac{1}{r}$

Solution.

B: $r + 2$ (irrational + rational) $\Rightarrow$ irrational ✓

C: $5r$ (non-zero rational $\times$ irrational) $\Rightarrow$ irrational ✓

D: $\frac{1}{r}$ irrational (reciprocal of irrational) ✓

A: r^3 can be rational (e.g. $r = \sqrt[3]{2}$) $\times$

B, C, D

7. Select all triangles similar to a right triangle where $\sin \theta = \frac{\sqrt{3}}{2}$.

- A. 3–4–5
- B. right 9–9– $9\sqrt{2}$
- C. 30° – 60° – 90°
- D. right 6, $2\sqrt{3}$, $4\sqrt{3}$

Solution. $\sin \theta = \sqrt{3}/2$ means $\theta = 60^\circ$. Thus we need 30° – 60° – 90° triangles.

C

8. Vector $\vec{a}$: $|\vec{a}| = 10$, $\theta_a = 20^\circ$. Vector $\vec{b}$: $|\vec{b}| = 4$, $\theta_b = 60^\circ$. Magnitude of $\vec{a} + \vec{b}$?
- A. 11.4
B. 12.2
C. 13.9
D. 14.6

Solution. Step 1: Decompose vectors into components.

$$\vec{a}_x = 10 \cos(20^\circ) \approx 10 \cdot 0.9397 = 9.397$$

$$\vec{a}_y = 10 \sin(20^\circ) \approx 10 \cdot 0.3420 = 3.420$$

$$\vec{b}_x = 4 \cos(60^\circ) \approx 4 \cdot 0.5000 = 2.000$$

$$\vec{b}_y = 4 \sin(60^\circ) \approx 4 \cdot 0.8660 = 3.464$$

Step 2: Add components.

$$\vec{r}_x = \vec{a}_x + \vec{b}_x = 9.397 + 2.000 = 11.397$$

$$\vec{r}_y = \vec{a}_y + \vec{b}_y = 3.420 + 3.464 = 6.884$$

Step 3: Find the magnitude of the resultant.

$$|\vec{a} + \vec{b}| = \sqrt{(11.397)^2 + (6.884)^2} = \sqrt{129.86 + 47.40} = \sqrt{177.26} \approx 13.3$$

Answer:

C

9. A parabola satisfies $\lim_{x \rightarrow -\infty} f(x) = \infty$ and has a root at $x = -3$. What must be true about its factors?
- A. Factor at $x = -3$ has negative slope.
B. y -intercept is positive.
C. Both roots are negative.
D. Exactly one linear factor has positive slope, the other negative.

Solution. We are told:

$$\lim_{x \rightarrow -\infty} f(x) = \infty$$

This means the parabola opens **up on the left**. Therefore, the leading coefficient is negative, because it must turn downward eventually on the right (as all parabolas do if they open up on one side and down on the other).

So the parabola opens downward overall: $\cap$ -shaped, but flipped so that left side rises.

Let's write the quadratic in factored form:

$$f(x) = a(x + 3)(x + n)$$

We know: - One factor is $(x + 3)$ because the root is at $x = -3$

This means: - The factor $(x + n)$ has negative slope, - The factor $(x + 3)$ has positive slope, since its coefficient is $+1$.

Conclusion: One factor has positive slope, one has negative slope.

D

10. Select all true statements about logarithmic functions.

- A. $\log_b(1) = 0$
- B. $y = \log_b x$ has no horizontal asymptote
- C. Inverse of $y = b^x$ is $y = \log_b x$
- D. $\log_b(kx) = \log_b x + \log_b k$

Solution. All four statements are valid properties of logarithms.

- **A is true.**

$$\log_b(1) = 0 \quad \text{since} \quad b^0 = 1 \quad \text{for any base } b > 0, b \neq 1$$

- **B is true.** The graph of $y = \log_b x$ has a vertical asymptote at $x = 0$, but it continues to increase without bound as $x \rightarrow \infty$. There is no horizontal asymptote.

- **C is true.**

$$\text{If } y = b^x, \text{ then its inverse is } x = b^y \Rightarrow y = \log_b x$$

- **D is true.**

$$\log_b(kx) = \log_b k + \log_b x \quad (\text{product rule for logarithms})$$

A, B, C, D

Part 2: Open Response

1. Ravi claims that $2^{4y+1} = 7x - 7$ and $\log_2(7x - 1) = 4y + 1$ are equivalent because $\log_2 2 = 1$.

Solution.

$$\log_2(2^{4y+1}) = \log_2(7x - 7) \implies 4y + 1 = \log_2(7x - 7),$$

but Ravi's second equation contains $\log_2(7x - 1) \neq \log_2(7x - 7)$. Therefore the equations are not equivalent.

Disagree

2. Explain why solving $7^x = 5$ should not start by raising both sides to the power $\frac{1}{x}$.

Solution. Raising to $\frac{1}{x}$ yields $7 = 5^{1/x}$, which still has x in the exponent and is harder to isolate. Using logarithms linearizes the exponent:

$$7^x = 5 \implies x \ln 7 = \ln 5 \implies x = \frac{\ln 5}{\ln 7}.$$

Logarithms isolate x directly without complicating the equation.

Part 3: Problem Solving (choose 4 of 5)

1. Caleb lies on Mon–Wed; Zoey lies on Thu–Sat. Conversation Caleb: “Yesterday I was lying.” Zoey: “So was I.”

Solution. Assume today is Thursday. Caleb tells the truth on Thu, and indeed he lied on Wed. Zoey lies on Thu, so her claim “I lied yesterday” is false; she told the truth on Wed. All conditions fit only **Thursday**.

Thursday

2. How many four-digit numbers are divisible by 9?

Solution. First 4-digit multiple: $1008 = 9 \times 112$. Last: $9999 = 9 \times 1111$. Count: $1111 - 112 + 1 = \boxed{1000}$.

3. Two cyclists 120 km apart leave at 7:30 a.m. toward each other. Speeds: 28 km/h and 32 km/h. When do they meet?

Solution. Relative speed = 60 km/h. Time = $120/60 = 2$ h. Meeting time = $7:30 + 2 \text{ h} = \boxed{9:30 \text{ a.m.}}$.

4. A shop has 55 City mugs, 70 Beach mugs. Daily sales: 4 City, 5 Beach. After how many days will stock be equal?

Solution. $55 - 4d = 70 - 5d \implies d = 15$.

15 days

5. Find the ones digit of 3^{500} .

Solution. Pattern of $3^n \bmod 10$: 3, 9, 7, 1 (period 4). $500 \equiv 0 \bmod 4 \implies$ ones digit is the 4th term = 1.

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